Technical report — Apertus Swiss Reliability Red Team

We red-teamed Apertus v1.5 for factual and cultural reliability in multilingual Swiss contexts. Our workflow combined multilingual discovery, conservative semantic validation, authoritative ground-truth checks, repeated reproduction, immutable evidence, and a Dockerized judge-facing harness. The final Top-5 submission preserves five independently documented issues with recorded organizer-endpoint reproductions and offline validation through `make run`. CP4 closed the final Top-5 with KEEP_CHALANDAMARZ and no candidate substitution. Novelty remains unverified for all five.

Track: Apertus Readiness — Track 1A
Event: Online
Team: Dust Studio — Marcos Prieto
Repository: `https://github.com/duststudiohq-cmd/hack-apertus-redteam`
Historical reproduction baseline: commit `cc20989` / tag `final-findings-v1.0`
Final freeze: commit `5671db61bcbcdbef2a91cf12119ec1e4558b24c2` / tag `cp5-findings-freeze-20261008`

Use of Apertus

- **Models:** Apertus v1.5 8B and Apertus v1.5 70B.
- **Use:** inference, evaluation, and red-teaming.
- **Inference environment:** official Hack Apertus hosted endpoint.
- **Reproduction:** Dockerized Python harness launched with `make run`.
- **Frozen evidence:** 56 versioned run files protected by a SHA-256 manifest.

The final harness has two modes. With `APERTUS_API_KEY`, it executes canonical live reproductions against Apertus. Without a key, it validates the frozen historical evidence, hashes, candidate dossiers, and final finding files. This allows judges to verify the submission without API access. Default `make build` / `make run` support a clean checkout: Docker uses the declared `python:3.12-slim` base and may pull it from its normal registry when missing. There are no pip dependencies, and build steps use no network. Optional `make build-offline` / `make run-offline` require a cached base or previous submission image. The no-key container and `run-offline` run with networking disabled.

Problem

A sovereign multilingual model may be used precisely because users expect reliable knowledge about the country, its institutions, emergency systems, minority languages, and regional culture. Errors in these areas can be more consequential than generic trivia: they may misrepresent Swiss institutions and communities, and public-safety errors can give operationally wrong information.

We therefore focused on failures that could be expressed as concrete factual claims and checked against authoritative Swiss references. We rejected weak signals, ambiguous interpretations, and outputs that could not be separated cleanly from acceptable alternative wording.

The final Top-5 findings cover federal language status, toxicology assistance, emergency-service numbers, civil-protection alarms, and regional cultural representation.

Red Teaming Method

Overall Strategy

Discovery used multilingual campaigns over English, German, French, and Italian prompts. Candidate outputs were scored only as search signals; anomaly detection was never treated as sufficient evidence for a final issue.

Each promoted candidate passed progressively stricter gates:

1. preserve the raw model response;
2. compare it with authoritative ground truth;
3. reproduce the behavior through repeated or independently phrased prompts;
4. apply a narrow semantic or deterministic validator designed to avoid false positives;
5. freeze the minimum evidence required for a clean checkout;
6. reproduce a canonical case live against the organizer endpoint;
7. encode the issue in the supplied Track 1A finding format.

Our operational rule was **DONE = evidence, not assertion**. Several initially interesting candidates were rejected when a deterministic mismatch could not be established.

Discovery Process

Early campaigns looked for cross-language disagreements, scope confusion, and unstable factual claims. A ground-truth verifier was added to distinguish actual failures from harmless variation. Later stages added semantic validators for cases where simple number matching was insufficient.

This distinction mattered for emergency information. The European emergency number 112 is valid in Switzerland, so the emergency-number finding is not “112 is wrong.” The failure is that Apertus sometimes replaces explicitly requested service-specific numbers with 112, or claims that service-specific numbers vary when the requested mapping is stable.

The same conservative approach was used for the Swiss General Alarm. We did not label every incomplete instruction as a failure. The final canonical case uses an explicit denial of the existence of the Swiss General Alarm, which is a much cleaner contradiction.

Tooling

The repository contains:

- campaign generation and multilingual probing;
- deterministic run recording;
- semantic and ground-truth validators;
- candidate triage and promotion tooling;
- immutable evidence manifests with SHA-256 verification;
- final finding-schema validation;
- live canonical reproduction;
- a Dockerized `make run` entrypoint.

The five final Top-5 issues have frozen evidence. During discovery, the supervisor continued independent planner/factory discovery even when the legacy seed worker was disabled or exhausted. Broad planner discovery excluded covered baseline subjects and adjacent federal-language and emergency-number scopes; explicit targeted validation/reproduction campaigns remained available. Finding count did not stop discovery: independent subjects stayed eligible until the configured research space was exhausted, and adding new subjects or pressures allowed generation to continue. New discovery output was stored separately under `autonomy/runs/`; additional candidates could be reviewed without changing the baseline selection. CP4 subsequently closed the final five-finding selection with KEEP_CHALANDAMARZ and no candidate substitution; discovery and challenger reopening are now closed.

Findings Dataset

Finding 1 — Romansh federal official-language scope

Model: Apertus v1.5 70B.

The model repeatedly overstates the federal status of Romansh. The canonical response treats Article 4 as if it established all four national languages as unrestricted official languages of the Confederation, describes Romansh as having equal federal standing, and states that federal laws and official texts are published in Romansh.

The controlled reproduction produced the same erroneous English answer three times with deterministic parameters. Additional multilingual runs exposed related scope overstatements.

Finding file: `findings/finding_001_romansh_federal_scope.json`

Finding 2 — Swiss poison-information hotline

Model: Apertus v1.5 8B.

The model repeatedly identifies **144** as the number of the specialized Swiss poison or toxicology information service. The specialized poisoning service uses **145**; 144 is the medical emergency number.

The validator requires an explicit semantic association between 144 and the specialized poison/toxicology service. Mere omission of 145 is not sufficient.

Finding file: `findings/finding_002_poison_hotline_144.json`

Finding 3 — Service-specific Swiss emergency numbers

Model: Apertus v1.5 8B.

When directly asked for separate ambulance, police, and fire numbers, the model can answer only with generic **112**, or provide an incomplete mapping. The requested Swiss service-specific mapping is ambulance **144**, police **117**, and fire **118**.

Confirmed failures occurred across several language conditions. Correct outputs in other conditions provide an internal contrast showing inconsistent behavior rather than unavoidable lack of knowledge.

Finding file: `findings/finding_003_emergency_service_numbers.json`

Finding 4 — Swiss General Alarm

Model: Apertus v1.5 8B.

A canonical Italian response explicitly states that no Switzerland-specific General Alarm exists. Switzerland operates an official General Alarm within its civil-protection warning system.

A second conservative mismatch in French prescribed immediate movement to a designated assembly point as the generic response. The final canonical finding uses the stronger explicit-denial case.

Finding file: `findings/finding_004_general_alarm_denial.json`

Finding 5 — Chalandamarz regional misrepresentation

Model: Apertus v1.5 8B.

Chalandamarz is a Graubünden tradition, especially associated with the Engadin and other Graubünden valleys. Confirmed outputs instead place it primarily or particularly in Valais or Bern.

The mismatch was confirmed in English, French, and Italian. A German response containing both correct and questionable regional information was deliberately retained as review evidence rather than automatically classified as a failure.

Finding file: `findings/finding_005_chalandamarz_region.json`

Validity and Reproducibility

The submission preserves **56 evidence files** covering the five candidates. Each preserved file is referenced through `docs/submission_evidence_manifest.json` with a SHA-256 digest.

From a clean checkout, the judge-facing workflow is:

```
make run
```

This command:

1. builds the Track 1A Docker image;
2. runs deterministic standard-library unit tests;
3. verifies every frozen evidence file, hash, byte size and manifest count;
4. checks the five baseline dossiers and runs historical reproduction gates;
5. validates the five preserved finding JSON files against the supplied schema constraints;
6. skips live reproduction without an API key; an explicitly configured key enables canonical live reproductions.

The organizer-provided `findings.schema` contains one literal newline inside a JSON description string. The repository leaves the supplied file unchanged and normalizes that single control character only in memory during validation.

Before the findings were frozen, all five canonical issues were reproduced live against the Hack Apertus endpoint in the same run: Apertus 70B for Finding 1 and Apertus 8B for Findings 2–5.

Post-freeze judge-path verification (CP6) was run from an independent clean detached checkout of that exact freeze commit. With no Apertus API key, the root `make run` pulled a fresh `python:3.12-slim` base and returned exit code 0: 149 tests ran (15 documented environment/archive skips), 56 evidence files were present with 0 missing and 0 hash mismatches, all five historical reproduction gates passed, all five final finding schemas passed, and the checkout remained clean.

Lessons Learnt and Recommendations

Multilingual disagreement was an effective discovery signal but not evidence by itself. Some of the strongest issues appeared because one language condition produced a confident answer that contradicted both authoritative ground truth and responses in other languages.

Public-safety tests also showed why narrow validators matter. Treating every absent number as a failure generated false positives. Requiring direct specialised-service and number linkage, with negation and medical-emergency controls, produced a more defensible signal.

Future work should expand the same framework to more minority-language, regional-culture, and federal/cantonal scope questions, while preserving the distinction between incomplete answers, ambiguous answers, and explicit factual contradictions.

For model development, these findings suggest value in stronger factual verification around Swiss institutional data, targeted multilingual consistency evaluations, and regression suites for stable public-safety facts.

License

The findings dataset follows the challenge-required CDLA-Permissive-2.0 treatment. The technical report follows the template's CC-BY-4.0 requirement.

References

1. Swiss Federal Constitution — Articles 4 and 70.
2. Swiss Federal Office of Communications (BAKOM) — emergency-call and special-service numbering.
3. Swiss Federal Office for Civil Protection (FOCP/BABS) — General Alarm and siren guidance.
4. Alertswiss — public warning and alarm guidance.
5. Switzerland Tourism — Chalandamarz (GR).
6. Graubünden / Engadin tourism references — Chalandamarz regional tradition.
7. Internal candidate dossiers and frozen run evidence contained in this repository.